

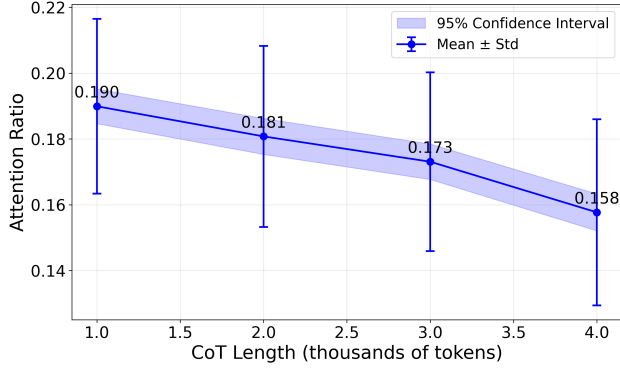


Figure 5. Attention ratio vs. CoT length (Qwen3-14B). Longer CoT sequences reduce relative attention to harmful instructions, weakening the safety check.

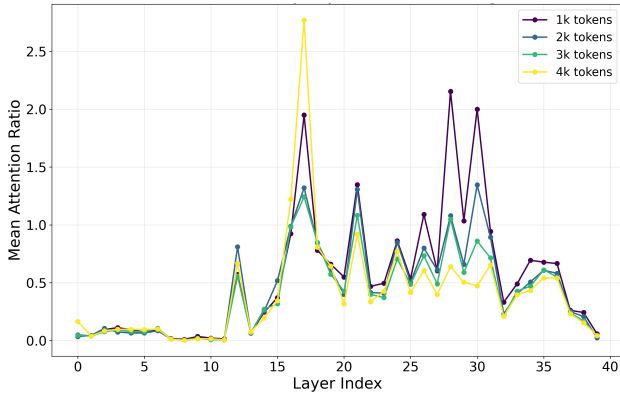


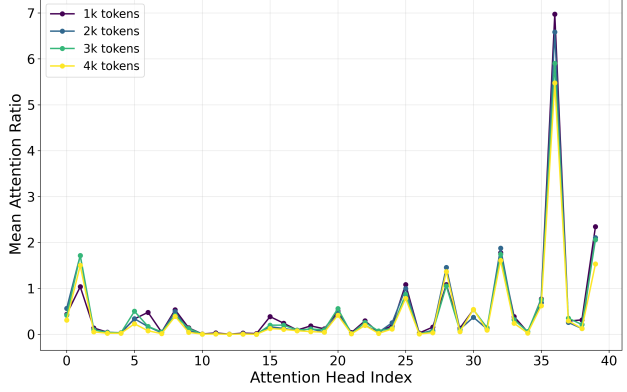
Figure 6. Layer-wise attention ratio across CoT lengths. During layers 15–35, longer CoT makes attention ratio decreases.

As shown in Figure 5, this ratio declines as CoT length increases, indicating that harmful instructions receive progressively less weight, thus diluting the safety mechanisms that detect and refuse unsafe requests. The layer-wise breakdown in Figure 6 pinpoints layers 15–35 as the locus where this decline is most pronounced; we therefore ablate heads in this range in the next Section 6.4.

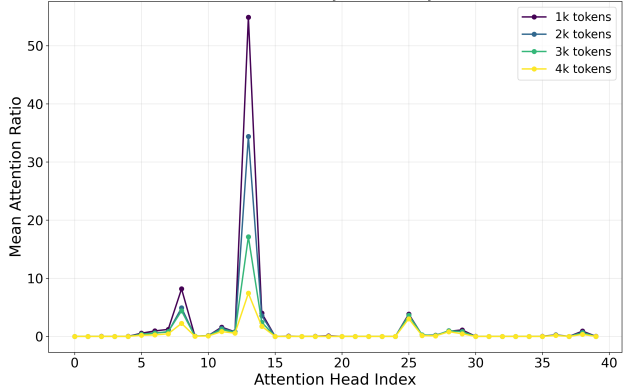
6.4. Attention-Head Intervention Experiments

We test whether the attention patterns identified above are merely correlational or actually causal for refusal. Building on the observation that layers 15–35 concentrate the strongest CoT-length-dependent attenuation, we select attention heads in this range and ablate them to see if refusal weakens correspondingly and thus have the alternative effect. This procedure allows us to identify the specific heads that drive the “longer CoT, weaker refusal” effect. In total, we manually select 60 specific heads that exhibit the ‘longer CoT, lower ratio’ pattern, such as those in Figure 7.

As shown in Figures 8 and 9, removing some of these heads flattens the distinction between harmful and harmless



(a) Layer 25 head-level attention ratio.



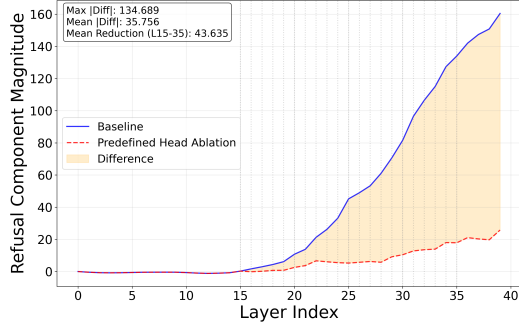
(b) Layer 30 head-level attention ratio.

Figure 7. Head-wise attention ratios for key layers (25, 30) as examples. Certain heads are primarily responsible for the CoT dilution effect. To select them, we focus on those exhibiting the ‘longer CoT, lower ratio’ pattern. For instance, in layer 30 (Figure 7b), heads (index) 13 and 8 clearly follow this trend and are therefore included.

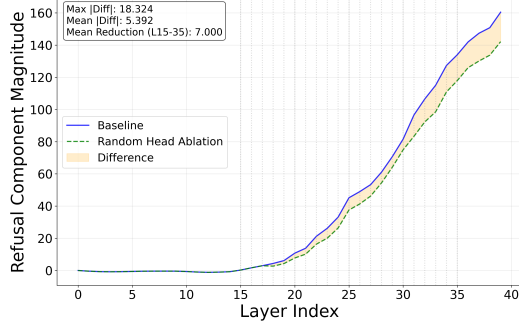
prompts, sharply reducing refusals. Targeted head removal is far more effective than random ablation, and among selected heads, front-layer heads (layer 15–23) have greater impact than later ones (layer 23–35). These causal interventions confirm that CoT Hijacking undermines a specific subnetwork of safety-critical heads, rather than degrading the model in a diffuse or incidental behavior.

6.5. Summary

Mechanistic evidence from both refusal components and attention maps shows that extended CoT length dilutes safety signals in late layers. Together, these findings suggest that jailbreaks succeed not only because refusal is a one-dimensional feature (Section 5), but also because its expression weakens as benign reasoning tokens dominate the context. We return to this synthesis in Section 7.

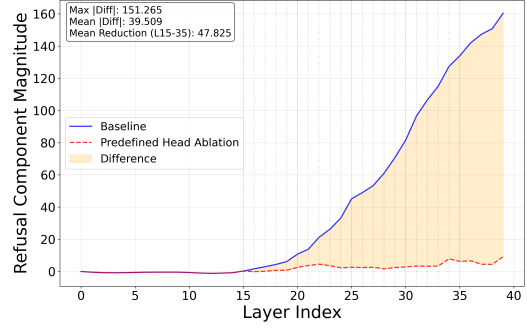


(a) Ablating 6 selected heads.

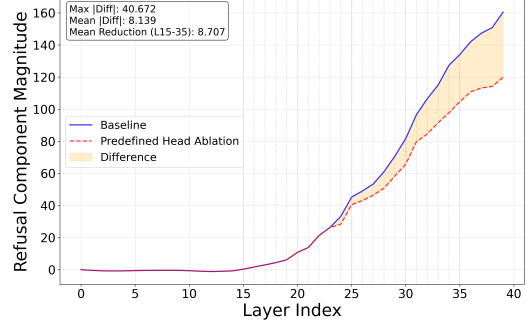


(b) Ablating 6 random heads.

Figure 8. **Selected vs Random ablation** (1k CoT, direct harmful query). Selected heads show stronger effect.



(a) Ablating 6 front-layer heads (layer 15–23).



(b) Ablating 6 deep-layer heads (layer 23–35).

Figure 9. **Front vs Deep heads ablation (among selected heads)**. Early-layer heads (layer15–23) play a stronger role in refusal control than deeper ones.

7. Results and Discussion

Our results establish that chain-of-thought reasoning, while improving accuracy, creates a new safety vulnerability. Across HarmBench, **CoT Hijacking achieves state-of-the-art attack success rates** (up to 99% on Gemini 2.5 Pro), outperforming prior jailbreak methods such as Mousetrap, H-CoT, and AutoRAN (Table 2). These results hold across multiple proprietary LRMs (Gemini, ChatGPT, Grok, Claude) and under consistent evaluation criteria, highlighting the generality of the attack.

Mechanistic evidence. Section 5 shows that refusal is mediated by a low-dimensional signal (the refusal direction) even in reasoning-augmented architectures. However, this signal is fragile: long chains of benign reasoning **dilute refusal activation**, while attention shifts away from harmful tokens. *Implications for scaling:* Our findings directly challenge the assumption that “more reasoning means more robustness” (Guan et al., 2024). Instead, scaling inference-time reasoning can exacerbate safety failures, especially in models explicitly optimized for long CoT. This calls into question alignment strategies that rely on shallow refusal heuristics without mechanisms that scale with reasoning depth.

Toward mitigation. The systematic nature of CoT Hijacking suggests that patching individual prompts is insufficient. Effective defenses may require deeper integration of safety into the reasoning process itself, such as monitoring refusal activation across layers, penalizing dilution, or enforcing attention to harmful spans regardless of reasoning length. We hope our findings motivate alignment strategies that are robust not only to short adversarial prompts but also to long chains of reasoning.

8. Conclusion

We introduced *CoT Hijacking*, a simple jailbreak attack against reasoning models. By padding harmful requests with long benign reasoning and a final-answer cue, we achieve high attack success rates across both open and proprietary LRMs. Through mechanistic analysis we found refusal components encode both the *strength* of safety checking in middle layers and the *outcome* of verification in later layers. Long CoT hijacking suppresses these signals, shifting attention away from harmful tokens and flattening refusal directions. Interventions on targeted attention heads confirmed their causal role, showing that hijacking undermines a specific safety subnetwork.

Acknowledgments

We thank Maksym Andriushchenko and James Oldfield for valuable feedback on earlier drafts of this work, and Antía García for assistance with visual design. We are grateful to the wider Oxford Martin AI Governance Initiative (AIGI) team for their continuous support. We also acknowledge OpenAI, Anthropic, and Martian for providing generous API credits and funding support.

Impact Statement

This paper identifies and analyzes a new vulnerability in large reasoning models, which we call *Chain-of-Thought Hijacking*. Because our method substantially increases the reliability of jailbreak attacks, it has **dual-use potential**. Our goal in releasing this work is to improve the security and robustness of deployed systems, not to enable misuse. To mitigate the risk of irresponsible application, we took the following steps: **Responsible Disclosure**. Before releasing this paper, we disclosed the vulnerability and provided logs, code, prompts and assistance to multiple major model developers, including OpenAI, Anthropic, Google DeepMind, and xAI. All groups acknowledged receipt, and several are actively evaluating mitigations. **Research-Only Usage**. We do not release any payloads, agentic exploit code, or system-targeted automation. The provided code is deliberately limited to benign demonstration prompts and controlled evaluation environments. **Motivation**. Our central motivation is to highlight that extended chain-of-thought reasoning—often assumed to enhance safety—can under certain conditions **reduce** a model’s refusal reliability. We believe that understanding and correcting this failure mode is essential for safely deploying reasoning-augmented systems. We emphasize that **this method should not be used for malicious purposes**. Disclosing vulnerabilities in a controlled and evidence-based manner is a standard practice in security research. Making these failure modes visible allows developers, policymakers, and the research community to **understand and mitigate** them. We believe that transparency, combined with early notification to affected developers, is the most responsible path toward improving the safety and reliability of AI systems, and we encourage others to follow similar responsible disclosure practices.

Reproducibility Statement

To ensure the reproducibility of our results, we introduce the experimental setup in Section 4. The code is released to facilitate relevant research.

References

- Arditi, A., Obeso, O., Syed, A., Paleka, D., Panickssery, N., Gurnee, W., and Nanda, N. Refusal in language models is mediated by a single direction. *Advances in Neural Information Processing Systems*, 37:136037–136083, 2024.
- Ball, S., Kreuter, F., and Panickssery, N. Understanding jailbreak success: A study of latent space dynamics in large language models. *arXiv preprint arXiv:2406.09289*, 2024.
- Chao, P., DeBenedetti, E., Robey, A., Andriushchenko, M., Croce, F., Schwag, V., Dobriban, E., Flammarion, N., Pappas, G. J., Tramer, F., et al. Jailbreakbench: An open robustness benchmark for jailbreaking large language models. *Advances in Neural Information Processing Systems*, 37:55005–55029, 2024.
- Chaudhary, M. and Barez, F. Safetynet: Detecting harmful outputs in llms by modeling and monitoring deceptive behaviors. *arXiv preprint arXiv:2505.14300*, 2025.
- Chen, J., Wang, X., Yao, Z., Bai, Y., Hou, L., and Li, J. Finding safety neurons in large language models. *arXiv preprint arXiv:2406.14144*, 2024.
- Denison, C., MacDiarmid, M., Barez, F., Duvenaud, D., Kravec, S., Marks, S., Schiefer, N., Soklaski, R., Tamkin, A., Kaplan, J., et al. Sycophancy to subterfuge: Investigating reward-tampering in large language models. *arXiv preprint arXiv:2406.10162*, 2024.
- Fatemi, M., Rafiee, B., Tang, M., and Talamadupula, K. Concise reasoning via reinforcement learning. *arXiv preprint arXiv:2504.05185*, 2025.
- Frankish, K. Dual-process and dual-system theories of reasoning. *Philosophy Compass*, 5(10):914–926, 2010.
- Fu, T., Sharma, M., Torr, P., Cohen, S. B., Krueger, D., and Barez, F. Poisonbench: Assessing large language model vulnerability to data poisoning. *arXiv preprint arXiv:2410.08811*, 2024.
- Fu, T., Gu, J., Li, Y., Qu, X., and Cheng, Y. Scaling reasoning, losing control: Evaluating instruction following in large reasoning models. *arXiv preprint arXiv:2505.14810*, 2025.
- Gao, L., Geng, J., Zhang, X., Nakov, P., and Chen, X. Shaping the safety boundaries: Understanding and defending against jailbreaks in large language models. In *Proceedings of the 63rd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pp. 25378–25398, 2025.